Project Proposal: Guided Super-Resolution for Boundary Refinement of TotalSegmentator Bone Segmentations in CT

1. Abstract

Automated tools such as TotalSegmentator segment many bones within seconds, but at a coarse resolution of 1.5 mm that falls short of the boundary precision available in the original CT. This project develops a post-processing step that refines such bone masks: a neural network receives the CT image together with the coarse mask and predicts a finer boundary (target resolution about 0.5 mm, to be confirmed). Earlier work has shown this for the femur only. We train on a public femur dataset (HFValid) and ask whether the learned refinement carries over to other bones without bone-specific training data. Because the network refines an existing mask rather than segmenting from scratch, it can learn local boundary behaviour (how CT intensity changes across the cortical surface), which makes transfer across bones plausible.

2. Background and motivation

Bone segmentations are used for surgical planning, finite-element strength models, implant design and fracture-risk assessment, where sub-millimetre boundary errors matter. General-purpose tools such as TotalSegmentator (Wasserthal et al.) cover many bones, but it is trained and applied at 1.5 mm resolution, so its boundaries are blocky and offset from the true cortical surface. For comparison, even a dedicated femur U-Net working at native CT resolution reaches an HD95 of roughly 1 mm (Bjornsson et al.), which illustrates how demanding sub-millimetre accuracy is.

Refining coarse masks with the help of the original CT has been demonstrated for the femur and, in related work, for the humerus (SCORE) and for brain MRI (Learn2Upscale). What is unclear is whether a refinement model learned on one bone transfers to others, since high-resolution ground truth for other bones is scarce.

Problem. Given a CT and a coarse TotalSegmentator bone mask, produce a finer boundary that is closer to expert ground truth, and quantify how well this transfers across bones. The project is limited to boundary refinement: gross segmentation errors (missing parts, wrong labels) are out of scope and will only be reported. We test a few representative bones and do not claim results for all bones.

3. Research questions

We pursue three research questions; the second is the main one.

- **RQ1 (in-domain benefit).** On femurs, does guided refinement improve on plain interpolation of the coarse mask, and is the coarse mask needed at all compared with a network that sees only the CT?
  - *H1:* The guided model reduces surface distance compared with interpolation and outperforms the CT-only model.
- **RQ2 (transfer, main question).** Does a femur-trained model improve TotalSegmentator boundaries on other bones (tibia and pelvis; vertebrae if time permits), and how does the gain relate to bone properties such as cortex thickness or density?
  - *H2:* The femur-trained model improves boundaries on other bones, but less so for thin-cortex bones.
- **RQ3 (what helps transfer).** Which design choices improve transfer: training with synthetic and perturbed coarse masks, or restricting the model to local context?
  - *H3:* Both improve transfer, with the context restriction possibly costing a little in-domain accuracy.

Since several configurations are compared, a failure to transfer across all of them is a stronger finding than the failure of a single model, and would motivate multi-bone training.

4. Data

Dataset	Role	Notes
HFValid (101 femur CTs with STL surfaces, CC BY 4.0)	Training and in-domain test	Spacing per subject in metadata; women >55, no implants
Fischer lower-body bone database (postmortem CT)	Cross-bone test: tibia, pelvis	Surfaces obtained by thresholding; CC BY-NC-SA; spacing to be verified
VerSe (optional)	Cross-bone test: vertebrae	Thin-slice scans only; labels partly model-assisted

The coarse inputs are obtained by running TotalSegmentator on each CT. All data are public and anonymised, and their licenses (including non-commercial terms) are respected. Two limitations apply: the HFValid cohort is narrow (women over 55, no implants), and on the other bones the ground truth is at best at native CT resolution, so sub-voxel tolerances are only meaningful on HFValid. Further public datasets (CTPelvic1K, CTSpine1K, RibSeg v2) are left for future work.

5. Approach

The project follows a standard pipeline: prepare data, establish references, train the models, and evaluate.

5.1 Data preparation

TotalSegmentator is run on every CT. The STL surfaces are converted to voxel masks on a common target grid, the CT is resampled to the same grid, and intensities are windowed in Hounsfield units (HU, the standard CT intensity scale). HFValid is split once at subject level into training, validation and test sets (roughly 70/10/20). The test set and all cross-bone data are never used for tuning or model selection.

5.2 Reference methods

- **Interpolation baselines (no learning, no CT).** Trilinear upsampling of the coarse mask, upsampling of the probability map followed by a threshold, and interpolation of a signed distance function. The best of the three is the main comparison for the guided model.
- **Oracle.** The same interpolation applied to ground truth downsampled to 1.5 mm. It shows how much of the error is due to resolution alone, as opposed to TotalSegmentator's own errors.
- **CT-only model.** An nnU-Net trained on femur CTs without any coarse mask. It learns to segment femurs from raw CT, which requires learning femur anatomy, so it is informative only in-domain: it shows whether the coarse mask is needed at all or whether a dedicated high-resolution femur segmenter would do. It is deliberately not used on other bones, where it would mainly measure anatomy recognition rather than boundary refinement.

5.3 Guided model

The guided model is an nnU-Net with two input channels, the CT on the target grid and the coarse TotalSegmentator mask upsampled to the same grid, trained on femurs only. The refined surface is extracted from the predicted probability map with marching cubes, which gives sub-voxel accuracy. Default nnU-Net settings are used, with no hyperparameter search beyond its self-configuration. Patch-based training and the default random flips and rotations are expected to favour local boundary features over whole-bone shape, but they do not guarantee it. Section 5.4 therefore tests this explicitly.

5.4 Generalization experiments (RQ3)

All configurations are trained on femurs only with the same split and training budget, and are evaluated in-domain and on the other bones, always on real TotalSegmentator outputs.

1. **Default.** The guided model of Section 5.3, trained on real TotalSegmentator masks.
2. **Perturbed coarse masks.** Additional training pairs with synthetic coarse masks made from the ground truth (downsampling to 1.5 mm, small random erosion and dilation, kept within the typical error of real TotalSegmentator masks). Only the input mask is perturbed; the target is always the unchanged fine ground truth.
3. **Restricted context.** The guided model with a small fixed nnU-Net patch size (for example 48 voxels per side, about 2.4 cm at 0.5 mm), so the network cannot see whole-bone shape and has to rely on local boundary appearance. A combination of 2 and 3 is trained if both help.

Optional extensions (only if time permits):

- **SDF regression.** Predict a truncated signed distance to the surface in a narrow band instead of a binary mask (custom training loop, truncated L1 loss). Distance to the surface is a local, shape-agnostic target and the output is continuous, which may transfer better.
- **Few-shot fine-tuning.** Fine-tune the femur model on a handful of subjects of a target bone to see how little target data is needed to close the remaining gap.

5.5 Reproducibility

Code is kept under version control with configuration files per run, fixed random seeds and split files, and pinned package versions. A README gives the commands needed to reproduce every table and figure.

6. Evaluation

All methods are evaluated on the same held-out femurs (in-domain) and on the other bones (cross-bone).

Metrics. Average symmetric surface distance (ASSD), HD95 and surface Dice at 0.5 mm and 1 mm tolerance, computed between the extracted predicted mesh and the ground-truth mesh to avoid voxel quantization.

Analysis. Results are reported as mean, standard deviation and median per metric, with paired per-subject differences between methods and a Wilcoxon signed-rank test as a sanity check. With small test sets, the results are treated as indications rather than firm claims. For RQ2 we compare the gain across bones and relate it to simple bone descriptors (e.g. mean HU near the boundary). For RQ3 the configurations are compared on in-domain accuracy and on the share of the in-domain gain retained on other bones. Selected cases, including failures, are inspected visually.

Expected outcome. Guided refinement most likely beats interpolation on femurs. Transfer should work best for bones with femur-like thick cortex and degrade for thin-cortex bones, with perturbed masks and restricted context partly mitigating this.

7. Risks and limitations

Risk	Mitigation
HFValid through-plane resolution is coarser than 0.5 mm	Check the metadata first; adjust the target resolution accordingly
TotalSegmentator is less reliable on the cropped, single-femur volumes in HFValid	Failure check; padding workaround; report excluded cases
Ground truth is threshold-like, so the model may learn the labelling protocol	Evaluate on a dataset with a different protocol (Fischer); state as limitation
Weak or low-resolution ground truth on other bones	Restrict claims; add visual comparisons

Outlook. Density-targeted augmentation, multi-bone training and additional datasets are natural extensions beyond the scope of this project.

8. Timeline and deliverables

The project is scoped for 5 ECTS, i.e. roughly 125 to 150 hours, or about 12 weeks at 10 to 12 hours per week.

- **Weeks 1-2:** data checks, TotalSegmentator inference, preprocessing, splits; quick check of the Fischer data
- **Week 3:** interpolation baselines and evaluation code, run on femur and cross-bone data
- **Weeks 4-6:** CT-only and guided models, in-domain evaluation
- **Weeks 7-8:** generalization experiments (perturbed masks, restricted context)
- **Weeks 9-10:** cross-bone evaluation and analysis
- **Weeks 11-12:** write-up and buffer; optional extensions only if ahead of schedule

Deliverables: a code repository with a script that refines a TotalSegmentator mask given the CT, and a final report with results.

9. Curriculum content

The project follows CRISP-DM: problem understanding (Sections 2-3), data understanding and preparation (Sections 4 and 5.1), modelling (5.2-5.4), evaluation (Section 6) and reuse (Section 8). Further curriculum methods applied:

Method	Where applied
Train/validation/test discipline, leakage control	Subject-level split, untouched cross-bone test sets
Data augmentation, overfitting	Section 5.4
Baselines, evaluation metrics, statistical testing	Sections 5.2 and 6
Reproducibility	Section 5.5

10. References

- Wasserthal et al., TotalSegmentator: robust segmentation of 104 anatomical structures in CT images, Radiology: AI 2023 (arXiv 2208.05868).
- Hu et al., Learning to Upscale 3D Segmentations in Neuroimaging, 2025 (arXiv 2505.21697).
- SCORE: refinement of TotalSegmentator segmentations on humerus CT, 2025 (arXiv 2511.02576).

- Bjornsson et al., Automated femur segmentation from CT using a deep neural network, 2021 (arXiv 2101.11742).